



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics

DJI Mavic Air Mobile Investigations

Android Phone

The Phone image and report are provided by [vtolabs](https://vtolabs.com)

Overview

- DJI Mavic Air
- Analyze main flight record *.txt
- Analyze video cache *.mp4
- Analyze detailed flight record *.dat
- Analyze cached pictures

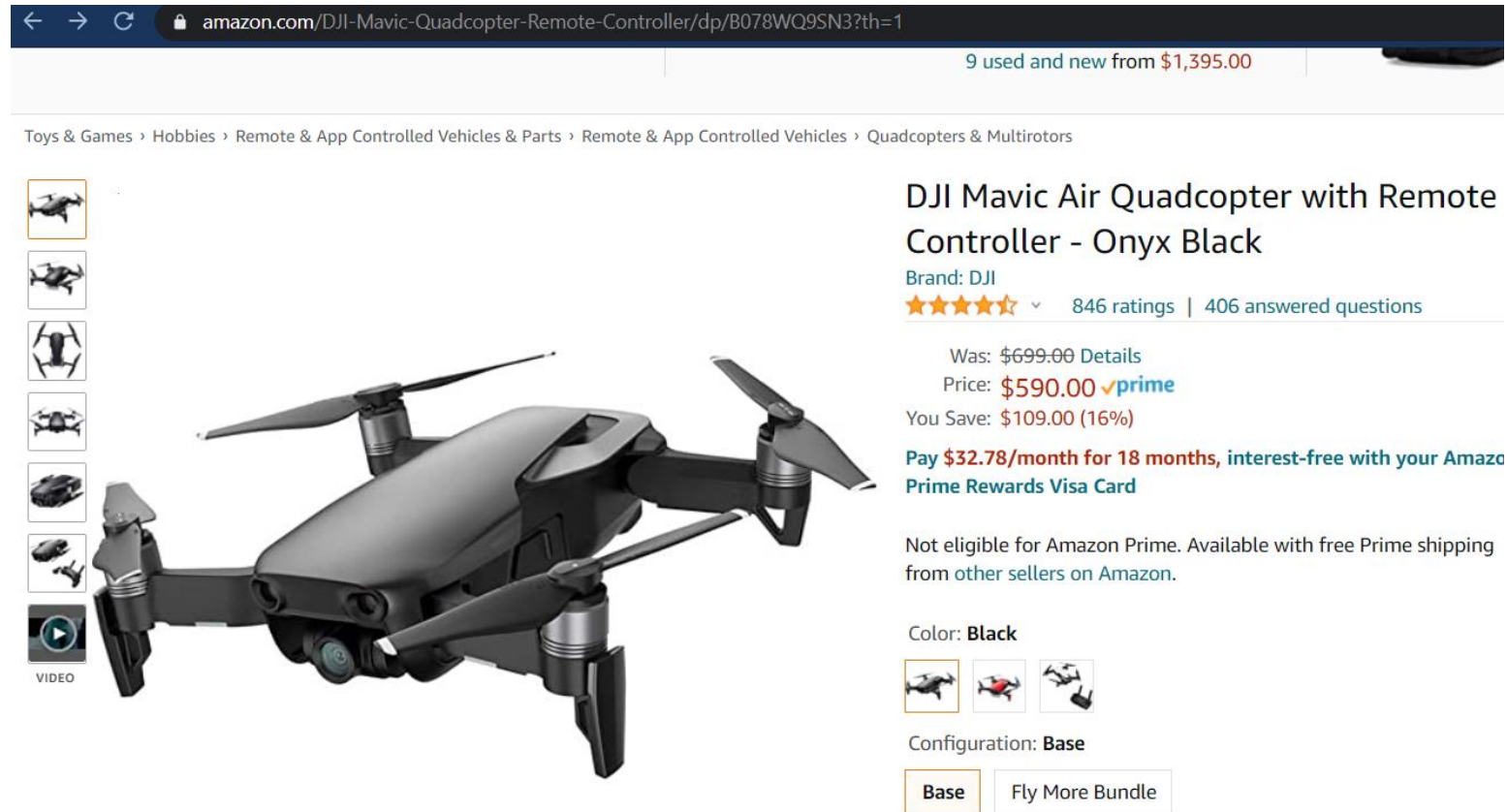
DJI Mavic Air

Content

- What is DJI Mavic Air?
- Mavic Air Specs
- Mavic Air Apps
- Forensic evidence types
- Data acquisition methods used for the lab

What is DJI Mavic Air?

- The DJI Mavic Air was released in 2018.
 - It is the 3rd model in the DJI Mavic series.
- DJI is a leading drone technology company
 - Privately owned
 - in China



The screenshot shows the Amazon product page for the DJI Mavic Air Quadcopter with Remote Controller - Onyx Black. The URL in the browser is amazon.com/DJI-Mavic-Quadcopter-Remote-Controller/dp/B078WQ9SN3?th=1. The page indicates that 9 used and new units are available from \$1,395.00. The breadcrumb trail shows the product is in the category of Toys & Games > Hobbies > Remote & App Controlled Vehicles & Parts > Remote & App Controlled Vehicles > Quadcopters & Multirotors. The product title is "DJI Mavic Air Quadcopter with Remote Controller - Onyx Black". The brand is DJI, and it has a 4.5-star rating from 846 ratings and 406 answered questions. The original price was \$699.00, and the current price is \$590.00, marked as a Prime deal. The savings are \$109.00 (16%). A financing option is available: "Pay \$32.78/month for 18 months, interest-free with your Amazon Prime Rewards Visa Card". A note states the product is not eligible for Amazon Prime but is available with free Prime shipping from other sellers. The color is Black, and the configuration is Base. There are buttons for "Base" and "Fly More Bundle". A vertical gallery of images on the left shows the drone from various angles and a video thumbnail.

amazon.com/DJI-Mavic-Quadcopter-Remote-Controller/dp/B078WQ9SN3?th=1

9 used and new from \$1,395.00

Toys & Games > Hobbies > Remote & App Controlled Vehicles & Parts > Remote & App Controlled Vehicles > Quadcopters & Multirotors

DJI Mavic Air Quadcopter with Remote Controller - Onyx Black

Brand: DJI

★★★★★ 846 ratings | 406 answered questions

Was: \$699.00 Details

Price: \$590.00 ✓prime

You Save: \$109.00 (16%)

Pay \$32.78/month for 18 months, interest-free with your Amazon Prime Rewards Visa Card

Not eligible for Amazon Prime. Available with free Prime shipping from other sellers on Amazon.

Color: Black

Configuration: Base

Base Fly More Bundle



Camera

Mavic Air Controller



MAVIC AIR Specs

 [Consumer](#) [Professional](#) [Enterprise](#) [Agriculture](#) [Explore](#) [Support](#) [Downloads](#) [Store](#)

Mavic Air [Specs](#) [Videos](#)

MAVIC AIR Specs

[Home](#) / [Products](#) / [Mavic Air](#) / [Specs](#)

Aircraft

Takeoff Weight	430 g
Dimensions	Folded: 168×83×49 mm (L×W×H) Unfolded: 168×184×64 mm (L×W×H)
Diagonal Distance	213 mm
Max Ascent Speed	4 m/s (S - mode ^[1]) 2 m/s (P - mode) 2 m/s (Wi-Fi mode)
Max Descent Speed	3 m/s (S - mode ^[1]) 1.5 m/s (P - mode) 1 m/s (Wi-Fi mode)

Gimbal

Mechanical Range	Tilt: -100° to 22° Roll: -30° to 30° Pan: -12° to 12°
Controllable Range	Tilt: -90° to 0° (default setting) -90° to +17° (extended)
Stabilization	3-axis (tilt, roll, pan)
Max Control Speed (tilt)	120°/s
Angular Vibration Range	±0.005°

Camera

Sensor	1/2.3" CMOS Effective Pixels: 12 MP
Lens	FOV: 85° 35 mm Format Equivalent: 24 mm Aperture: f/2.8 Shooting Range: 0.5 m to ∞
ISO Range	Video: 100 - 3200 (auto) 100 - 3200 (manual) Photo: 100 - 1600 (auto) 100 - 3200 (manual)
Shutter Speed	Electronic Shutter: 8 - 1/8000s
Still Image Size	4:3: 4056×3040 16:9: 4056×2280
Still Photography Modes	Single shot HDR Burst shooting: 3/5/7 frames Auto Exposure Bracketing (AEB): 3/5 bracketed frames at 0.7EV Bias Interval: 2/3/5/7/10/15/20/30/60 s Pano: 3×1: 42°×78°, 2048×3712 (W×H) 3×3: 119°×78°, 4096×2688 (W×H) 180°: 251°×88°, 6144×2048 (W×H) Sphere (3×8+1): 8192×4096 (W×H)
Video Resolution	4K Ultra HD: 3840×2160 24/25/30p 2.7K: 2720×1530 24/25/30/48/50/60p FHD: 1920×1080 24/25/30/48/50/60/120p HD: 1280×720 24/25/30/48/50/60/120p
Max Video Bitrate	100Mbps
Supported File System	FAT32
Photo Format	JPEG/DNG (RAW)
Video Format	MP4/MOV (H.264/MPEG-4 AVC)

Supported SD Cards

Supported SD Cards

microSD
Class 10 or UHS-1 rating required

Recommended microSD Cards

16G
Sandisk Extreme
Kingston 16GB SDHC-I SDCAC/16GB
32G
Sandisk Extreme V30
Sandisk Extreme V30 A1
Sandisk Extreme V30 Pro A1
64G
Sandisk Extreme V30 A1
Sandisk Extreme Pro V30 A1
128G
Sandisk Extreme V30 A1
Sandisk Extreme Plus V30 A1

APP

Video Transmission System

Name

Live View Quality

Enhanced Wi-Fi

DJI GO 4

Remote Controller:
720p@30fps
Smart Device:
720p@30fps
DJI Goggles:
720p@30fps

Latency (depending on environmental
conditions and mobile device)

170 - 240 ms

DJI Apps

dji.com/downloads/djiapp

Search

Top Downloads

DJI Apps ^

- DJI Store
- DJI Fly
- DJI Mimo
- DJI Ronin
- DJI Virtual Flight
- DJI Pilot
- DJI Pilot PE
- DJI GO 4
- DJI GO
- DJI GS Pro (iPad)
- DJI Ronin Assistant
- RoboMaster

Software v

Products v

DJI Apps



DJI Store

The DJI Store app lets you effortlessly shop for DJI products and enjoy exclusive discounts.



DJI Fly

The DJI Fly app interface was designed to be simplified and ultra-intuitive. Compatible for Mavic Mini, Mavic Air 2, DJI Mini 2



DJI Mimo

As an app dedicatedly designed for DJI handheld gimbals, DJI Mimo offers HD live video view, intelligent modes that hardly found on other handheld stabilizers, empowering you to



DJI Ronin

Get impressive filming results with DJI RS 2, DJI RSC 2, Ronin-S, and Ronin-SC



DJI Pilot

The DJI Pilot app supports the latest DJI enterprise drones.



DJI Pilot PE

DJI Pilot PE is a customized version of the DJI Pilot flight control app, which is exclusively deployed with FlightHub on a private cloud server.



DJI GO 4

Capture the world from above. Compatible with the Mavic Series, Spark, Phantom 4, and more.



DJI GO

Capture the world from above. Compatible with the Phantom 3 Series, Matrice Series , OSMO Series, and more.



DJI Ronin Assistant



RoboMaster

DJI GO 4--For drones since P4

← → ↻ play.google.com/store/apps/details?id=dji.go.v4&hl=en_US&gl=US

Google Play Search package name: dji.go.v4 🔍

Apps Categories ▾ Home Top charts New releases

My apps
Shop
Games
Kids
Editors' Choice

Account
Payment methods
Play Points New
My subscriptions



DJI GO 4--For drones since P4

DJI TECHNOLOGY CO., LTD Video Players & Editors ★★★★★ 57,818

Everyone

Contains Ads
This app is available for your device

You can share this with your family. [Learn more about Family Library](#)

 Add to Wishlist

Install

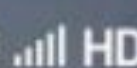


In Flight (GPS)

Position



20



HD

70%



18:25

Auto ISO

100

SHUTTER

1000

EV

+0.0

WB

Auto

1080P/30

49:22

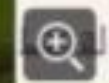
CAPACITY

100%

AF-C/MF

AE

AE



0 200m



D 301 m

H 39 m

H.S 29.1 km/h

V.S 0.0 m/s

307 m



0:48



FIX

DJI GO 4--For drones since P4

- Compatible with the **Mavic** Series, Spark, Phantom 4, and more.
- Features:
 - Near Real-time HD Image Transmission
 - Camera Settings Adjustment
 - Updated playback interface
 - Updated Editor with improved UI
 - More templates and music tracks in Editor
 - Convenient video downloading, editing and sharing
 - Integrated live streaming
 - Near real-time flight data recording

Forensic evidence types

Main Evidence Types	Other Evidence Types
Device (mobile/Drone) name Flight Timestamps/duration Flight speed Flight mode (e.g., Motor Started, AutoTakeoff, Autolanding, P-GPS, Manual) On Air Timestamps/duration Longitude, Latitude, Altitude Audio, Image, Video Distance To Home Flight Status/Mode Notification Battery name/information and Avoidance Weather (from timestamps)	Air pressure Altitude Acc X, Y, Z (Accelerometer) Gyro X, Y, Z (Gyroscope) Velocity North, East, Down, Valid Compass X, Y, Z Satellite Numbers Pitch, Roll, Yaw RC Input Pitch, Roll, Yaw RC Status RC Input Throttle Vision Height

P-GPS: The aircraft is using GPS for positioning.

Data Acquisitions

- The internal storage of the Mavic Air can be acquired by attaching the drone while powered on to your forensic workstation and imaging it as a standard USB drive.

	Description	Size	Note
1.	External microSD card	32GB	Aircraft Main Circuit Board
2.	eMMC component #1	16GB	Aircraft Main Circuit Board
3.	eMMC component #2	128Mb	Controller Main Circuit Board
4.	Mobile application	182MB/14.2GB	iOS and Android

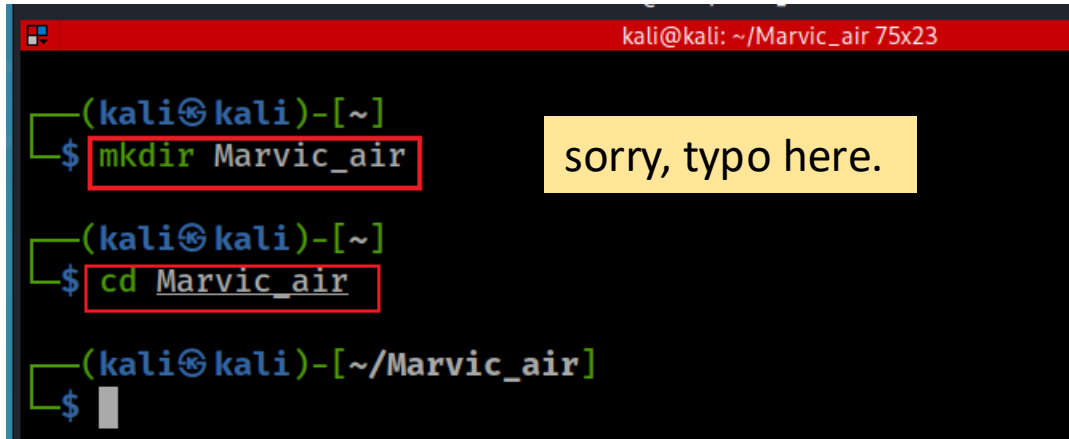
Table : Flight Record Files

Device	Location	Desc	Example
Phone	Dji/dji.go.v4/FlightRecord	Main flight record (Motor on->Landed only)	DJIFlightRecord_2018-06-19_[14-50-34].txt
	Dji/dji.go.v4/DJI_Record	Video Cache (Takeoff->Landed only)	2018_06_19_14_50_39.mp4
	Dji/dji.go.v4/FlightRecord/MCDatFlightRecords	Detailed flight record (PowerOn->PowerOff)	(20)18-06-19-02-47-38_FLY057.DAT
Mavic	Internal storage (DJI Assistant 2 to export)	Drone flight	DJI_ASSISTANT_EXPORT_FILE_2018_06_19_14_50_39.DAT

Analyze main flight record *.txt

Motor on -> Landed

Create a working folder



```
kali@kali: ~/Marvic_air 75x23

(kaliⓈkali)-[~]
$ mkdir Marvic_air

(kaliⓈkali)-[~]
$ cd Marvic_air

(kaliⓈkali)-[~/Marvic_air]
$
```

Terminal window showing the creation and navigation to a folder named Marvic_air. The window title is "kali@kali: ~/Marvic_air 75x23". The prompt is "(kaliⓈkali)-[~]". The command "mkdir Marvic_air" is entered and executed. The prompt is "(kaliⓈkali)-[~]". The command "cd Marvic_air" is entered and executed. The prompt is "(kaliⓈkali)-[~/Marvic_air]". The prompt is "\$".

sorry, typo here.

Download Android logical

https://github.com/yaukacheung/csci352/blob/main/A5/Android_Logical.zip

Download zip

```
kali@kali: ~/Marvic_air 106x23
(kali@kali)-[~/Marvic_air]
$ wget https://www.dropbox.com/s/0d21nfd3wxfvfls/Android_Logical.zip
--2021-05-02 16:50:37-- https://www.dropbox.com/s/0d21nfd3wxfvfls/Android_Logical.zip
Resolving www.dropbox.com (www.dropbox.com)... 162.125.6.18, 2620:100:601c:18::a27d:612
Connecting to www.dropbox.com (www.dropbox.com)|162.125.6.18|:443... connected.
```

Verify md5

```
(kali@kali)-[~/Marvic_air]
$ ls -l
total 405428
-rw-r--r-- 1 kali kali 415154789 May  2 17:01 Android_Logical.zip

(kali@kali)-[~/Marvic_air]
$ md5sum Android_Logical.zip
4f7623911818e8dc1ddceee4e26875be Android_Logical.zip ←
```

unzip

```
(kali@kali)-[~/Marvic_air]
$ unzip Android_Logical.zip

(kali@kali)-[~/Marvic_air]
$ ls
Android_Logical ← Android_Logical.zip
```

File Structure of DJI Go 4

```
kali@kali: ~/Marvic_air 107x48
(kali@kali)-[~/Marvic_air]
$ tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots

13 directories, 0 files
```

- Recorded flights
- Definition of a flight
 - Motors ON-> Landed
- Evidence collected
 - Almost all types
 - Not include video

How to view a flight router information?

(option 1: manual)

Find the flight starting and ending time

```
kali@kali: ~/Marvic_air 107x48

(kali@kali)-[~/Marvic_air]
$ tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
├── DJIFlightRecord_2018-06-19_[14-50-34].txt ← This is a local/motor ON. Need to find the UTC time.
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT
    └── 18-06-19-02-59-05_FLY058.DAT

1 directory, 3 files

(kali@kali)-[~/Marvic_air]
$ stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Size: 555776      Blocks: 1088      IO Block: 4096   regular file
Device: 801h/2049d Inode: 401982   Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2018-06-19 12:55:40.000000000 -0400
Modify: 2018-06-19 12:55:40.000000000 -0400 ← last modified time / a flight ending time
Change: 2021-05-03 15:24:15.379562225 -0400
Birth: 2021-05-03 15:24:15.375560227 -0400

Landed time in UTC: 2018-06-19 16:55:40
```

Find the flight device name

```
(kali㉿kali)-[~/Marvic_air]
$ file Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34\].txt
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt: data

(kali㉿kali)-[~/Marvic_air]
$ xxd -s -200 Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34\].txt
00087a38: 3239 4596 bb46 2680 10f6 e274 c7ff ff64 29E..F8....t...d
00087a48: 726f 6e65 666f 2e2e 2e2d 4d61 7669 6320 ronefo...-Mavic
00087a58: 4169 7200 0000 0000 0000 0000 0000 0030 Air.....0
00087a68: 4b31 4446 3331 3342 4434 5052 3100 0000 K1DF313BD4PR1...
00087a78: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00087a88: 0000 0000 0000 0000 0000 0000 0000 0030 .....0
00087a98: 4b34 4145 4251 4133 3430 3044 5400 0002 K4AEBQA3400DT...
00087aa8: 0402 1000 0000 0000 0000 0000 0100 0000 .....
00087ab8: 0000 0000 0000 000f 3ed6 0d40 42ac 1967 .....>..@B..g
00087ac8: 3db7 0c41 44e6 124d 38f1 0e44 41a1 1a4e =..AD..M8..DA..N
00087ad8: 3eb4 0540 42a6 194c 35b7 0d42 43a6 154c >..@B..L5..BC..L
00087ae8: 38b5 0d41 4ea5 184f 38b6 0043 43a5 184c 8..AN..08..CC..L
00087af8: 22b7 0c40 42a7 fcff "...@B...
```

device name

this .txt format has special format,
we **can't** interpreter the data
without a parser/tool.

How to view a flight router information?

(option 2: www.phantomhelp.com)

https://www.phantomhelp.com/LogViewer/Upload/

ali Tools Kali Docs Kali Forums NetHunter Offensive Security Exploit-DB GHDB MSFU

JUST RELEASED The new DJI Air 2S drone is now available!

PhantomHelp START HERE ACCESSORIES & TIPS GUIDES FORUM NEWS STORE DRONE D

Upload Your Flight Log

Click the "Browse..." button below to select and upload your **DJI GO TXT** flight log (e.g. "DJIFlightRecord_2016-02-12_[15-33-10].txt") or your **Litchi CSV** flight log.

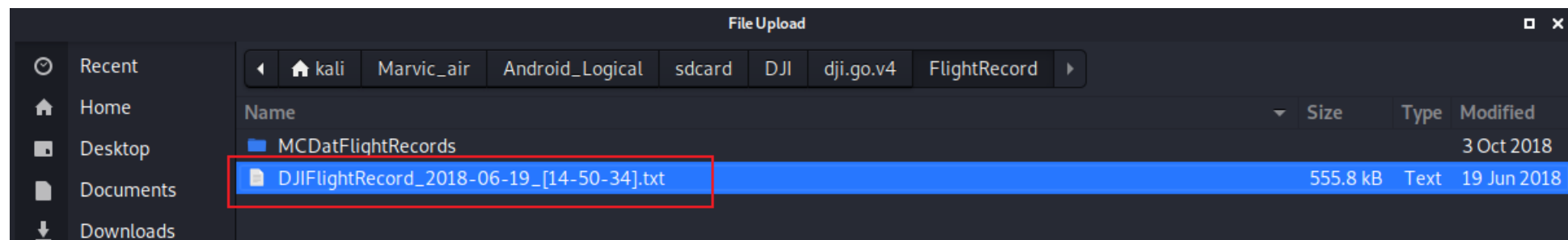
✓ I'm not a robot


reCAPTCHA
[Privacy](#) - [Terms](#)

Browse...

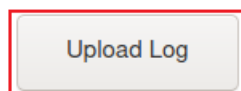
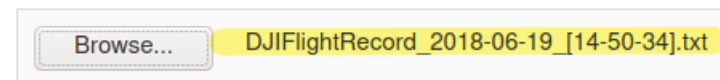
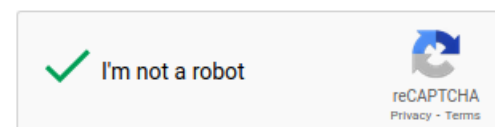
No file selected.

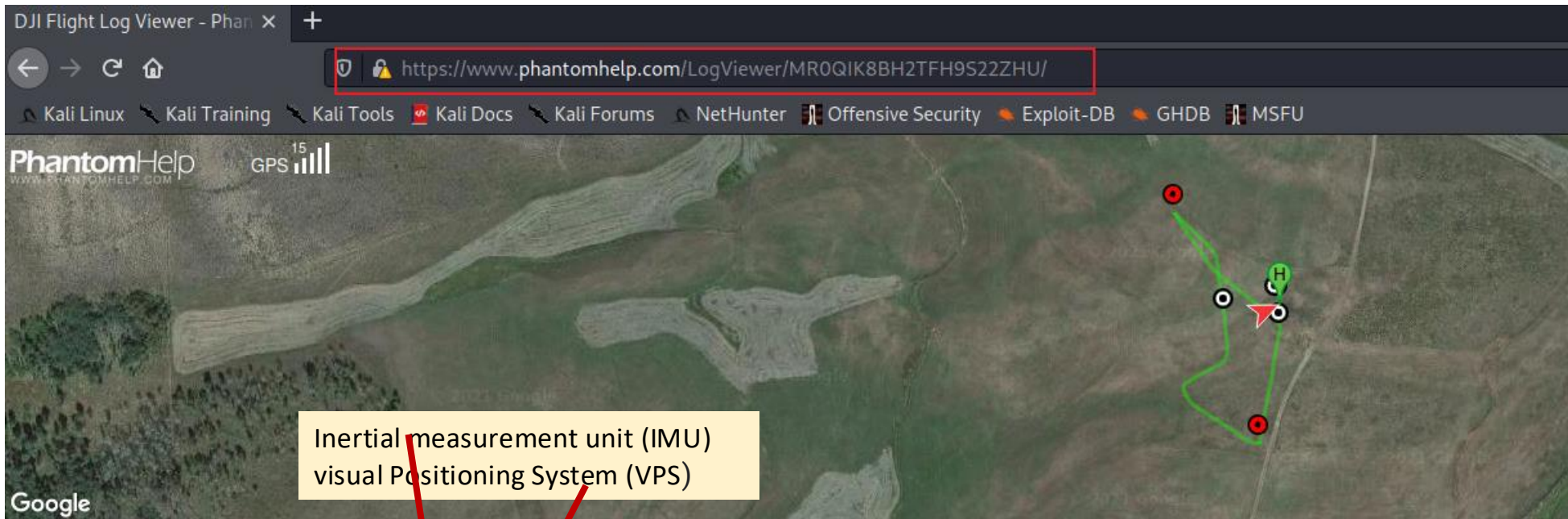
Upload Log



Upload Your Flight Log

Click the "Browse..." button below to select and upload your **DJI GO TXT** flight log (e.g. "DJIFlightRecord_2016-02-12_[15-33-10].txt") or your **Litchi CSV** flight log.



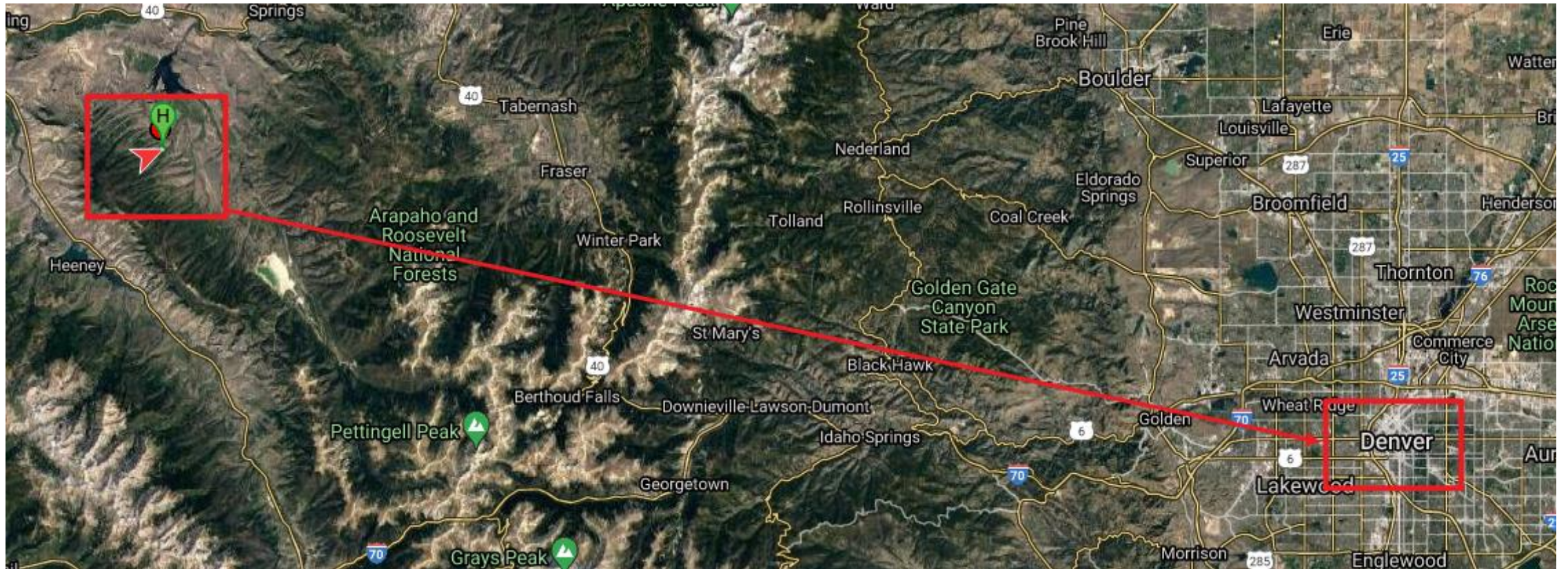


[Download KML](#) | [Download CSV](#) | [Download Verbose CSV](#) | [Download Flight Log](#)

Time	Flight Mode	GPS	IMU Altitude	VPS Altitude	Speed	Home Distance	Battery	Battery Voltage	Cell 1	Cell 2	Cell 3	Cell Deviation	Message
0m 0.2s	Starting Motors	15 satellites	0 ft	0.3 ft	0 mph	3.1 ft	95%	12.697 V	4.229 V	4.233 V	4.235 V	0.006 V	
0m 1.5s	Auto Takeoff	15 satellites	0 ft	0.3 ft	0 mph	3.2 ft	95%	12.692 V	4.225 V	4.232 V	4.235 V	0.01 V	
0m 2.1s	Auto Takeoff	16 satellites	0 ft	0.3 ft	0 mph	3.2 ft	95%	12.692 V	4.225 V	4.232 V	4.235 V	0.01 V	
0m 2.2s	Auto Takeoff	16 satellites	0 ft	0.3 ft	0.2 mph	0 ft	95%	12.692 V	4.225 V	4.232 V	4.235 V	0.01 V	Home Point Updated
5m 5s	Auto Landing	19 satellites	-6.6 ft	0.3 ft	0 mph	2.2 ft	68%	11.712 V	3.909 V	3.905 V	3.898 V	0.011 V	
5m 5.4s	P-GPS	19 satellites	0 ft	0.3 ft	0 mph	2.2 ft	68%	11.712 V	3.909 V	3.905 V	3.898 V	0.011 V	

5 min 6 sec

Find the location of the flight



How to verify the flight starting and ending time?

```
kali@kali: ~/Marvic_air 107x48

(kali@kali)-[~/Marvic_air]
$ tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
├── DJIFlightRecord_2018-06-19_[14-50-34].txt
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT
    └── 18-06-19-02-59-05_FLY058.DAT

1 directory, 3 files
```

This is Colorado local time (UTC -6 hours)
UTC time: 2018-06-19 20:50:34

Did the flight last 4 hours 5 minutes and 6 second?

```
(kali@kali)-[~/Marvic_air]
$ stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Size: 555776      Blocks: 1088      IO Block: 4096   regular file
Device: 801h/2049d Inode: 401982    Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2018-06-19 12:55:40.000000000 -0400
Modify: 2018-06-19 12:55:40.000000000 -0400
Change: 2021-05-03 15:24:15.379562225 -0400
Birth: 2021-05-03 15:24:15.375560227 -0400
```

last modified time / a flight ending time

UTC time: 2018-06-19 16:55:40

dji flight records txt mavic timestamp issue



All

News

Shopping

Videos

Images

More

Settings

Tools

About 32,000 results (0.61 seconds)

<https://forum.dji.com/thread-183553-1-1>

Flight Logs Showing wrong date | DJI FORUM

Mar 11, 2019 — - Has anyone else noticed this with the Smart Controller and the **Mavic 2 Pro**? All the settings are correct as far as Date & Time go in the controller ...

[40 posts](#) · Hi All, Did an 11-minute flight today (3/11/19) and when I got home and checked the ...

Missing: ~~txt~~ | Must include: [txt](#)

<https://forum.dji.com/thread-174491-1-1>

Timestamp on video is incorrect,while pictures are fine | DJI ...

Dec 11, 2018 · 17 posts · 12 authors

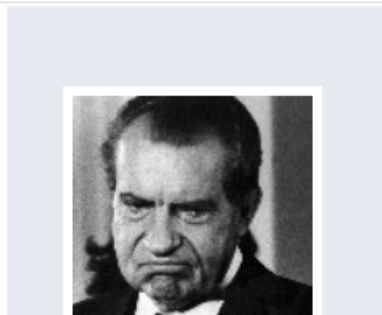
Hi,On **Mavic 2 Pro**,latest firmware,the media creation **timestamp** on the video files ... The metadata actually shows the correct time within the MP4/MOV and if you **record** a caption/SRT, it also shows the correct time. ... so I suspect this is an **issue** with the Android version of the **DJI Fly** app. ... **Bold Text Color**.

Missing: ~~txt~~ | Must include: [txt](#)

video timestamp incorrect

[Home](#) / [Products](#) / [Phantom](#) / [Phantom 3 Pro/Adv](#)

[4020](#) [23](#) [2016-](#)



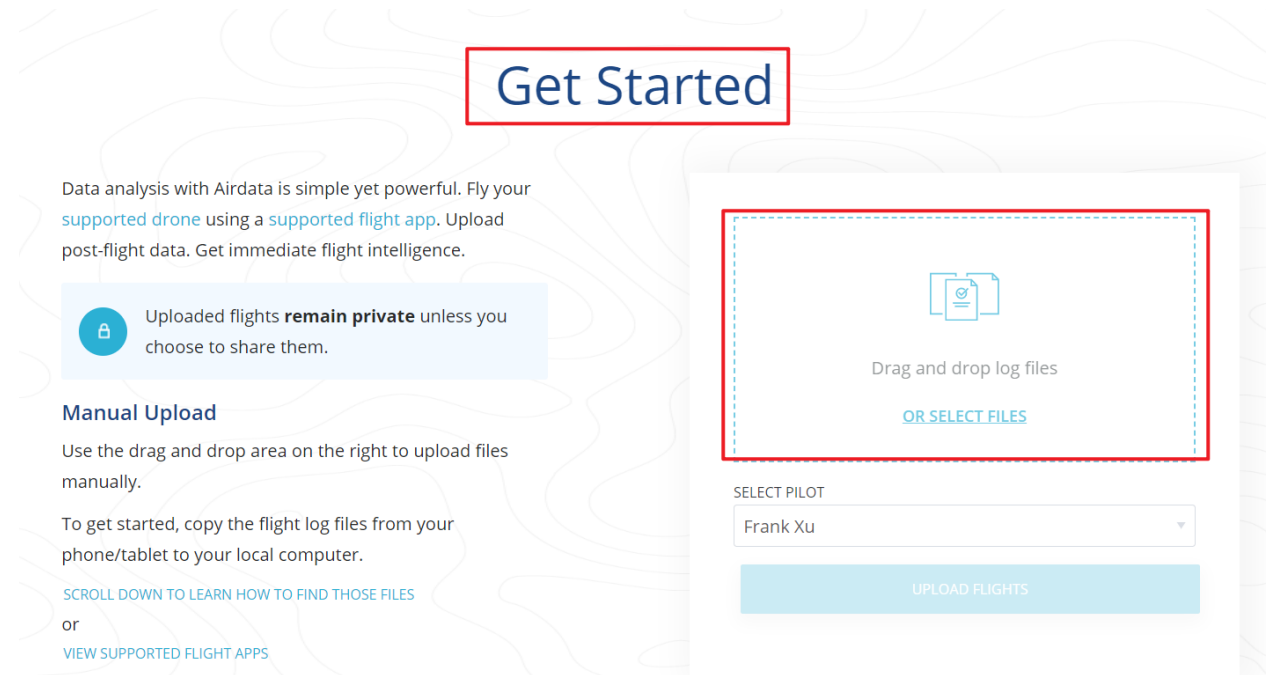
I have noticed that after I import the photos and videos off the SD card, the timestamp for the videos is off by 4 hours. The photos are correct. For example, I was flying around 7 PM last night. The videos indicate they were taken around 3 PM, but the photos correctly indicate 7 PM. Is there some setting where I can get the clock for the videos set correctly?

it seems the time zone did not set up correctly.

How to view a flight router information?

(option 3: airdata.com)

- Provide more information than options 1 and 2
- Free for personal use
- Need to register



is the timestamp correct?

[Metric / Imperial](#)
[Settings](#)

Overview

Details

Equipment

Notifications

Large Map

Jun 19th, 2018 12:50PM [Edit](#)

GENERAL

POWER

SENSORS

CONTROLS

WEATHER

MEDIA

Jun 19th, 2018 12:50PM (-06:00)

Plane Name
dronefo-Mavic Air

Flight Air Time
05m 02s

Takeoff Battery
95% 12.7v

Landing Battery
68% 11.7v

Mavic Air/Android
DJI 4.2.16

Map

Satellite

Google

Map Data

Terms of Use

Report a map error

Total Mileage
1,459 ft

Max Distance
325 ft

Max Altitude
93.5 ft

Max Speed
6.51 mph

Max Bat Temp
N/A

Tips: **1**
Warnings: **6**

Download: [KML](#) [GPX \(?\)](#) [CSV](#) [Original](#)

Add tag

[Metric / Imperial](#)
[Settings](#)

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Large Map



GENERAL

Associated Pilots:



PILOT-IN-COMMAND



United States

POWER

SENSORS

CONTROLS

WEATHER

MEDIA

Take Off Location: **39.961198,-106.216468**
Above Sea Level: **8127.8 ft**

Address: **Unnamed Road, Parshall, CO 80468, USA**
[Edit Address](#)

Last Known Location: **39.961197,-106.216476**
at log time of **05m 05s**
and altitude of **0.0 ft**

Log Duration: **05m 05s** (time in the air and on the ground)

Air Duration: **05m 02s** (time in the air only)

[Metric / Imperial](#)
[Settings](#)

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GENERAL

POWER

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WEATHER

MEDIA

Associated Drone: [dronefo-Mavic Air](#)
Type **Mavic Air**

Associated Batteries: [Bat-MavicAir-3400DT](#)



GENERAL



Review notification highlights from the flight.
Click "HD Flight Player" to view detailed flight info.

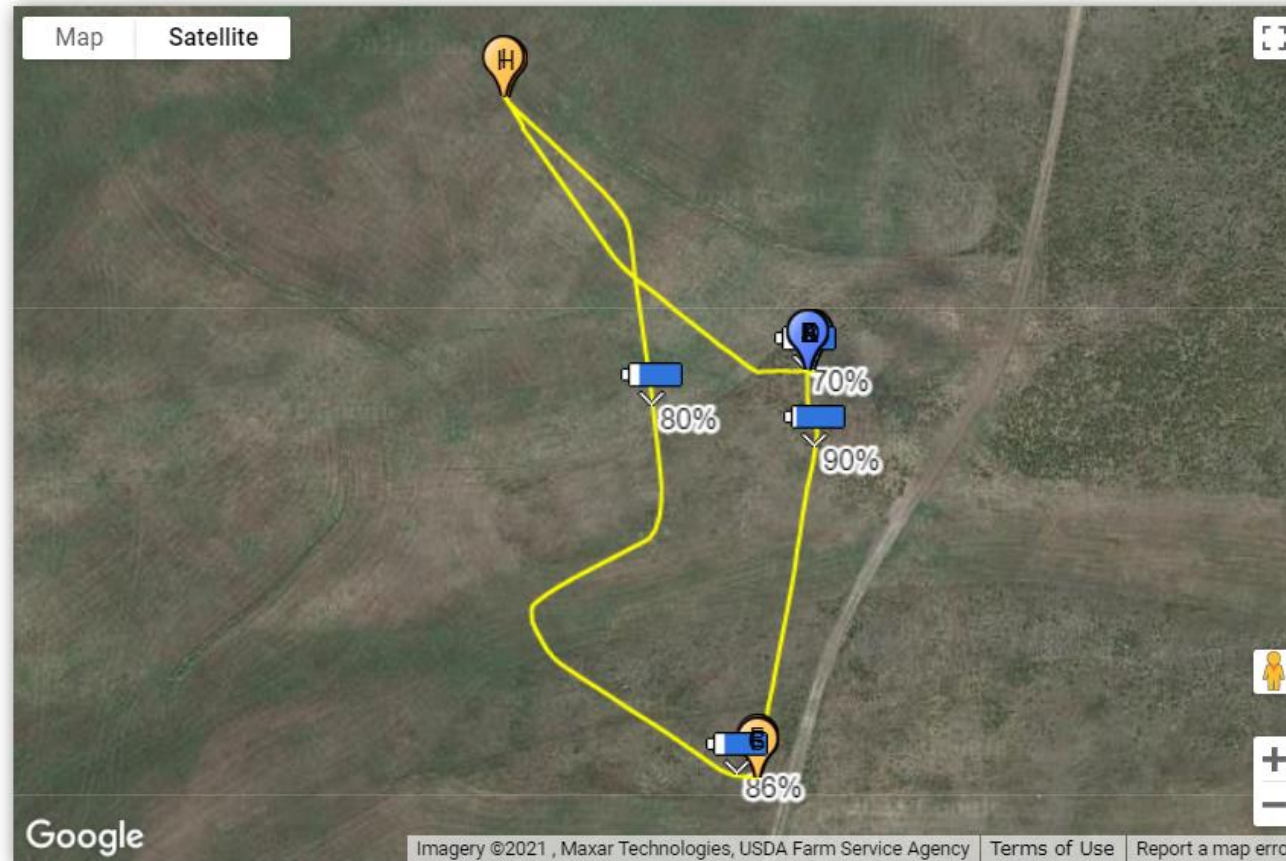
POWER

SENSORS

CONTROLS

WEATHER

MEDIA



NEW!

[Download notifications KML](#)

	Flight time	Altitude	Home Dist	Type	Notification
A	<u>00m 00s</u>	0.0 ft	0 ft	Mode	Mode changed to Motors Started
B	<u>00m 00s</u>	0.0 ft	0 ft	Tip	✓ <u>Setting new Return-To-Home altitude to 30m (98 ft).</u> ✓ <u>Data Recorder File Index is 57.</u> ✓ <u>Setting new Maximum Flight Altitude to 50m (164 ft).</u>
C	<u>00m 01s</u>	0.0 ft	0 ft	Mode	Mode changed to AutoTakeoff

	Flight time	Altitude	Home Dist	Type	Notification	Start recording Mode
A	<u>00m 00s</u>	0.0 ft	0 ft	Mode	Mode changed to Motors Started	
B	<u>00m 00s</u>	0.0 ft	0 ft	Tip	✓ <u>Setting new Return-To-Home altitude to 30m (98 ft).</u> ✓ <u>Data Recorder File Index is 57.</u> ✓ <u>Setting new Maximum Flight Altitude to 50m (164 ft).</u>	
C	<u>00m 01s</u>	0.0 ft	0 ft	Mode	Mode changed to AutoTakeoff	
D	<u>00m 10s</u>	12.8 ft	1 ft	Mode	Mode changed to P-GPS	
	<u>00m 54s</u>	14.4 ft	62 ft		90% Battery	
E	<u>01m 31s</u>	73.2 ft	322 ft	Warning	Maximum flight distance reached. Adjust the distance in MC Settings if required	
F	<u>01m 33s</u>	73.5 ft	324 ft	Low Risk	 <u>Weak signal. Avoid blocking the antennas and keep the antennas parallel to and facing toward the aircraft during flight</u>	
G	<u>01m 39s</u>	72.2 ft	324 ft	Warning	High wind velocity. Ensure the aircraft remains within your line of sight and fly with caution	
	<u>01m 44s</u>	72.5 ft	325 ft		86% Battery at maximum distance	
	<u>02m 42s</u>	81.7 ft	128 ft		80% Battery	
H	<u>03m 16s</u>	82.3 ft	323 ft	Warning	Maximum flight distance reached. Adjust the distance in MC Settings if required	
I	<u>03m 31s</u>	82.7 ft	324 ft	Warning	High wind velocity. Ensure the aircraft remains within your line of sight and fly with caution	
	<u>04m 33s</u>	-3.3 ft	4 ft		70% Battery	
J	<u>04m 57s</u>	-3.0 ft	3 ft	Mode	Mode changed to AutoLanding	
K	<u>04m 57s</u>	-3.0 ft	3 ft	Warning	Obstacle Sensing will be disabled when aircraft is landing. Fly with caution	
L	<u>05m 05s</u>	0.0 ft	2 ft	Mode	Mode changed to P-GPS	End recording GPS Position Mode



GENERAL

POWER

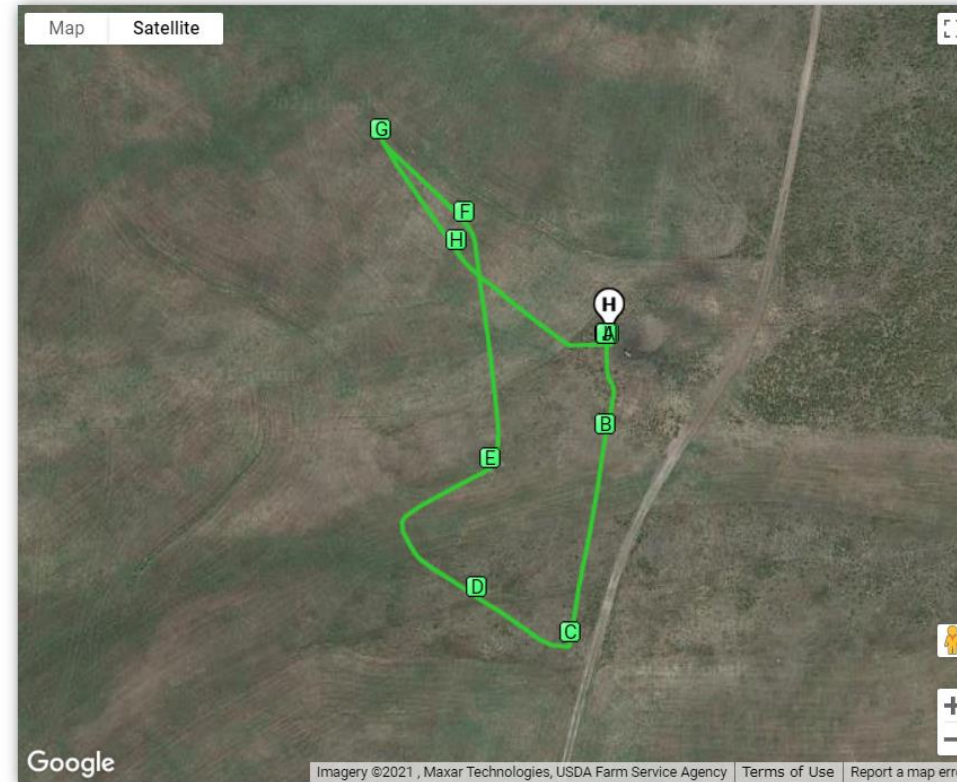
SENSORS

CONTROLS

WEATHER

MEDIA

Signal Strength Map - **green** is good signal, **orange** is fair, **red** is poor and **purple** is very poor. It calculates signal strength based on the connection to the remote - it searches for signal loss based on minor signal interruptions:



[Download color coded Signal Strength KML](#)

	Flight time	Altitude	Home Distance	Minor Signal Errors	Calculated Signal
A	00m 30s	13.1 ft	1 ft	0	100%
B	01m 00s	14.8 ft	96 ft	0	100%
C	01m 30s	73.2 ft	319 ft	0	100%
D	02m 00s	81.4 ft	304 ft	0	100%
E	02m 30s	82.0 ft	183 ft	0	100%
F	03m 00s	81.7 ft	202 ft	0	100%
G	03m 30s	82.7 ft	325 ft	0	100%
H	04m 00s	64.3 ft	192 ft	0	100%
I	04m 30s	-3.3 ft	4 ft	0	100%
J	05m 00s	-4.3 ft	2 ft	0	100%

[Metric / Imperial](#)
[Settings](#)

GENERAL

POWER

SENSORS

CONTROLS

WEATHER

MEDIA

Ground Weather

Kp Index

InFlight Wind

Wind Map

Alt. Profile



Temperature

70.5 °f



Clear

24% Cloud Cover

Visibility

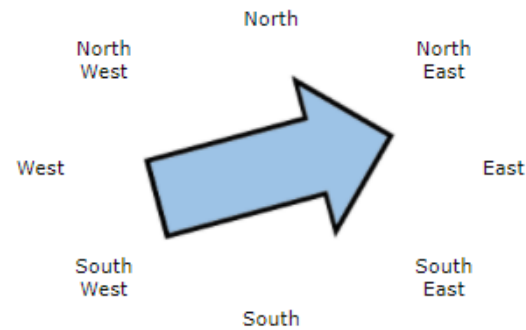
Excellent

7 Miles

Wind Speed

12.1 mph

Wind Direction



Direction: **258°**

Humidity

18 %

 05:37am Sunrise

 08:38pm Sunset



Waxing Crescent Moon
48% visible

Feels like : **70.5** °f
Dew Point : **25.2** °f

Pressure: Not Reported

Rain Rate : **0.00** in/h
Chance : **0** %

Analyze Video Cache *.mp4

Take off -> Landed

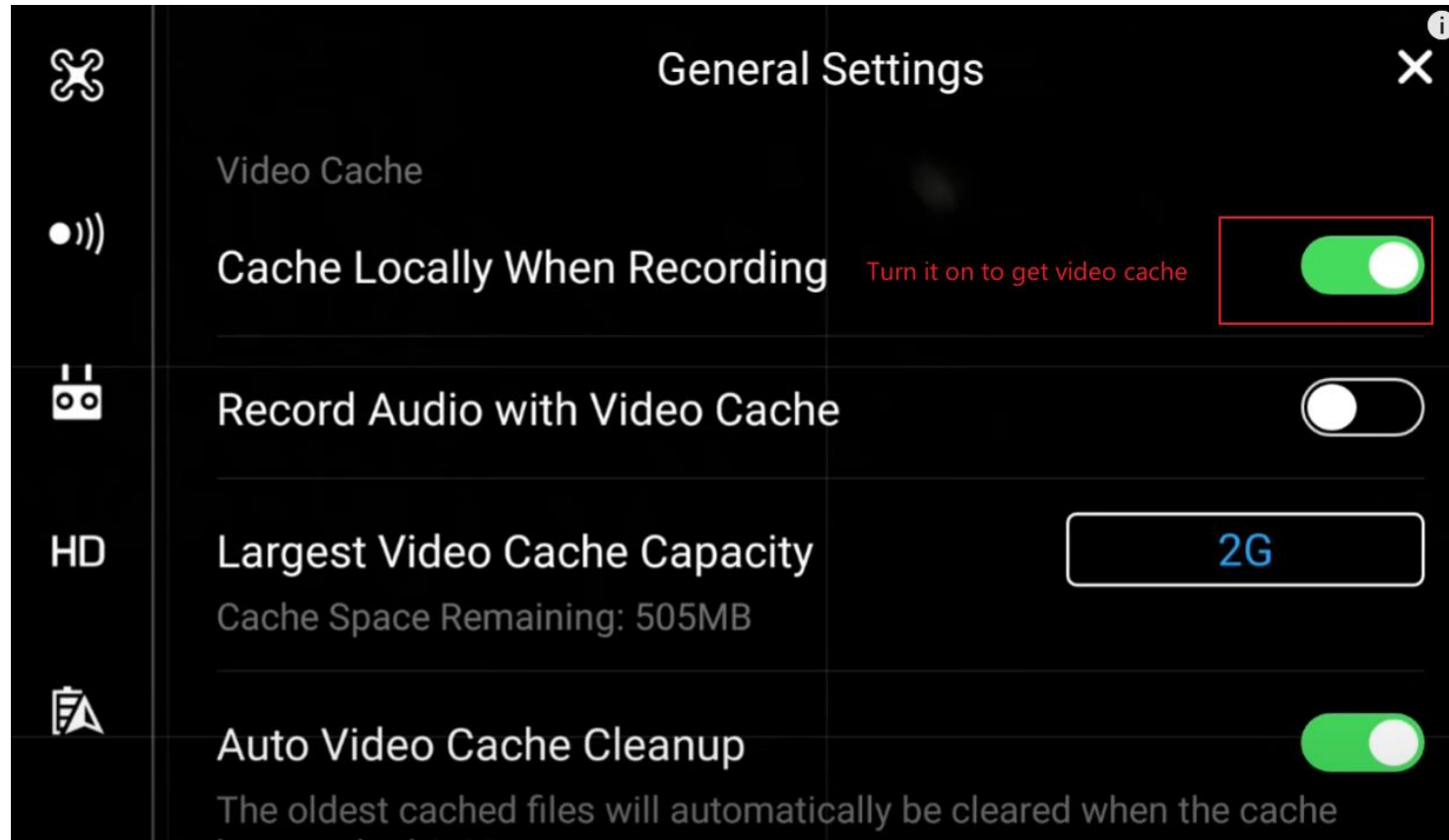
Where are flight videos (.mp4) saved?

```
kali@kali: ~/Marvic_air 107x48
(kali@kali)-[~/Marvic_air]
$ tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots

13 directories, 0 files
```

- Recorded flight videos
- Definition of a flight video
 - TakeOff -> Landed
 - Not recording until **Altitude >0**

Video Cache setting



When was the flight video (.mp4) recorded?

Show folder/files
and start
recording time

```
kali@kali: ~/Marvic_air 107x40
(kali@kali)-[~/Marvic_air]
$ tree -L 2 Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/ --inodes
Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/
├── [ 401821] 2018_06_19_14_50_39.info
├── [ 401822] 2018_06_19_14_50_39.mp4 ←
├── [ 401823] analytics
│   ├── [ 401824] 2018_06_19_14_50_39.map
│   └── [ 401825] 2018_06_19_14_50_39.tlv
└── 1 directory, 4 files
```

UTC time: 2018-06-19 16:50:349
note: the original timestamp in the
file name is 4 hours off

Show recording
ends time

```
(kali@kali)-[~/Marvic_air]
$ stat Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
File: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
Size: 57791765      Blocks: 112880      IO Block: 4096      regular file
Device: 801h/2049d  Inode: 401822       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2021-05-03 10:50:14.020215962 -0400
Modify: 2018-06-19 12:55:42.000000000 -0400 ←
Change: 2021-05-03 10:45:48.063250441 -0400
Birth: 2021-05-03 10:45:47.771396436 -0400
```

UTC time: 2018-06-19 16:55:402

	Flight time	Altitude	Home Dist	Type	Notification	Start recording Mode
A	<u>00m 00s</u>	0.0 ft	0 ft	Mode	Mode changed to Motors Started	
B	<u>00m 00s</u>	0.0 ft	0 ft	Tip	✓ <u>Setting new Return-To-Home altitude to 30m (98 ft).</u> ✓ <u>Data Recorder File Index is 57.</u> ✓ <u>Setting new Maximum Flight Altitude to 50m (164 ft).</u>	
C	<u>00m 01s</u>	0.0 ft	0 ft	Mode	Mode changed to AutoTakeoff	
D	<u>00m 10s</u>	12.8 ft	1 ft	Mode	Mode changed to P-GPS	
	<u>00m 54s</u>	14.4 ft	62 ft		90% Battery	
E	<u>01m 31s</u>	73.2 ft	322 ft	Warning	Maximum flight distance reached. Adjust the distance in MC Settings if required	
F	<u>01m 33s</u>	73.5 ft	324 ft	Low Risk	⚠ <u>Weak signal. Avoid blocking the antennas and keep the antennas parallel to and facing toward the aircraft during flight</u>	
G	<u>01m 39s</u>	72.2 ft	324 ft	Warning	High wind velocity. Ensure the aircraft remains within your line of sight and fly with caution	
	<u>01m 44s</u>	72.5 ft	325 ft		86% Battery at maximum distance	
	<u>02m 42s</u>	81.7 ft	128 ft		80% Battery	
H	<u>03m 16s</u>	82.3 ft	323 ft	Warning	Maximum flight distance reached. Adjust the distance in MC Settings if required	
I	<u>03m 31s</u>	82.7 ft	324 ft	Warning	High wind velocity. Ensure the aircraft remains within your line of sight and fly with caution	
	<u>04m 33s</u>	-3.3 ft	4 ft		70% Battery	
J	<u>04m 57s</u>	-3.0 ft	3 ft	Mode	Mode changed to AutoLanding	
K	<u>04m 57s</u>	-3.0 ft	3 ft	Warning	Obstacle Sensing will be disabled when aircraft is landing. Fly with caution	
L	<u>05m 05s</u>	0.0 ft	2 ft	Mode	Mode changed to P-GPS	End recording GPS Position Mode

video recording
started: + 05 sec

recording

video recording
ended: + 02 sec

```
kali@kali: ~/Marvic_air 106x23

(kali⊗kali)-[~/Marvic_air]
$ sudo apt-get install mediainfo
[sudo] password for kali:
```

```
kali@kali: ~/Marvic_air 107x40

(kali⊗kali)-[~/Marvic_air]
$ mediainfo Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
General
Complete name          : Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
Format                 : MPEG-4
Format profile         : Base Media
Codec ID               : isom (isom/iso2/avc1/mp41)
File size              : 55.1 MiB
Duration               : 4 min 0 s
Overall bit rate      : 1 919 kb/s
Writing application    : Lavf56.15.102

Video
ID                     : 1
Format                : AVC
Format/Info           : Advanced Video Codec
Format profile        : High@L3.1
Format settings       : CABAC / 1 Ref Frames
Format settings, CABAC : Yes
```

4 minutes

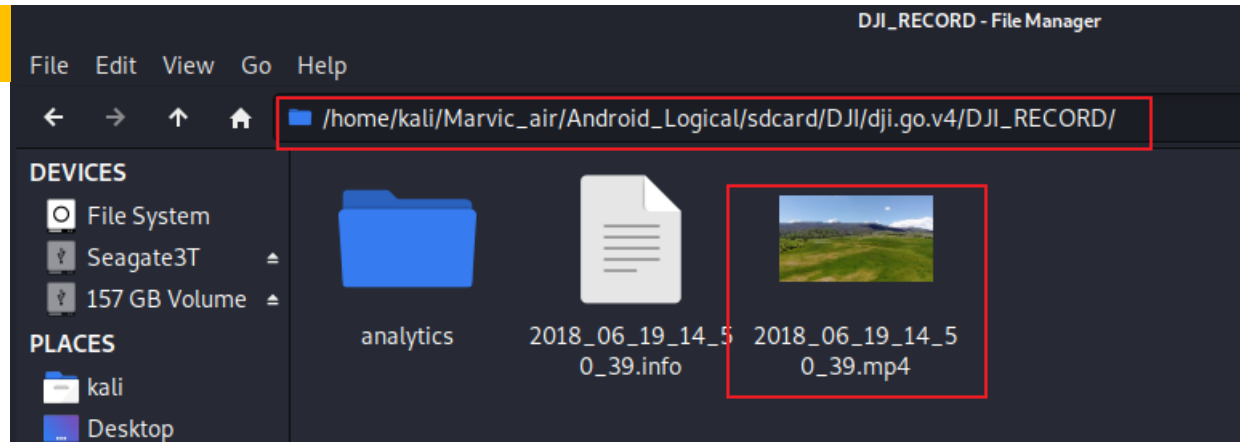
```
kali@kali: ~/Marvic_air 106x23

(kaliⓈkali)-[~/Marvic_air]
$ sudo apt install libimage-exiftool-perl
Reading package lists... Done
Building dependency tree
```

```
kali@kali: ~/Marvic_air 107x40

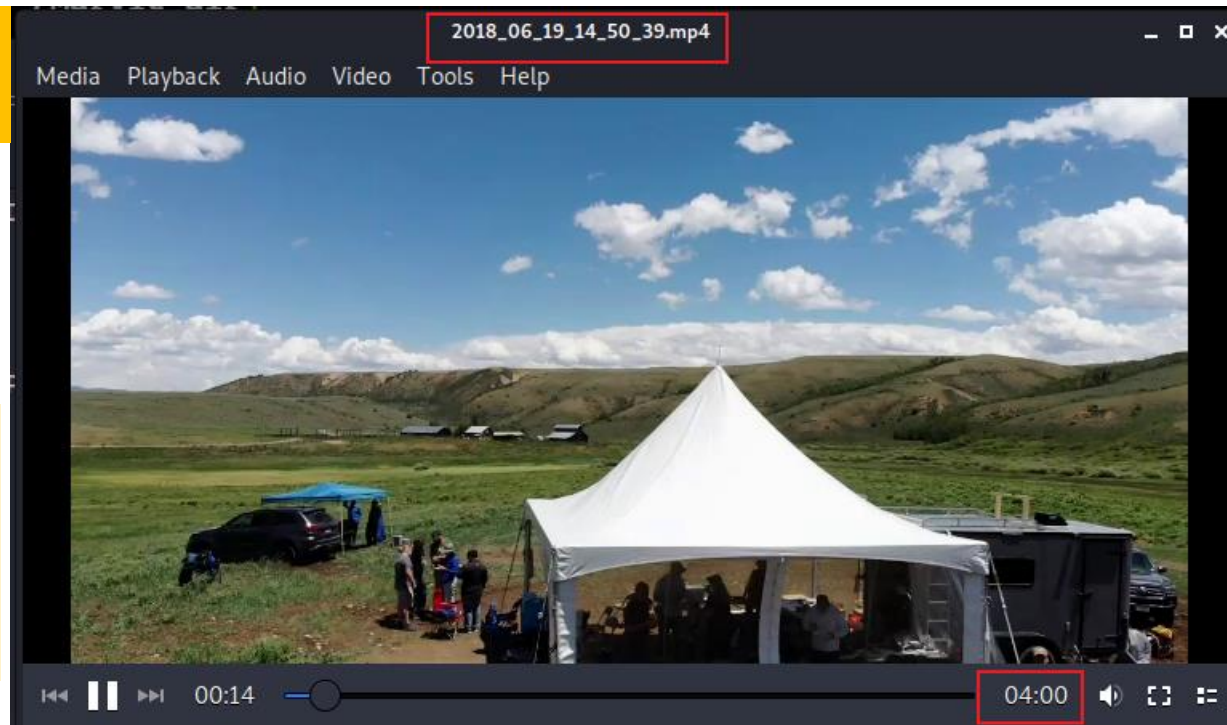
(kaliⓈkali)-[~/Marvic_air]
$ exiftool Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
ExifTool Version Number      : 12.16
File Name                    : 2018_06_19_14_50_39.mp4
Directory                   : Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD
File Size                    : 55 MiB
File Modification Date/Time  : 2018:06:19 12:55:42-04:00
File Access Date/Time       : 2018:06:19 12:55:42-04:00
File Inode Change Date/Time  : 2021:05:03 10:45:48-04:00 ← I unzip the zip file
File Permissions             : rw-r--r--
File Type                   : MP4
File Type Extension         : mp4
MIME Type                   : video/mp4
Major Brand                  : MP4 Base Media v1 [ISO 14496-12:2003]
Minor Version                : 0.2.0
Compatible Brands            : isom, iso2, avc1, mp41
```

Play recorded flight



The video lasts 4 minutes

note: the video access time will be change after playing the video. Use *stat* command to verify



View information related to the video

```
(kali@kali)-[~/Marvic_air]
$ tree -L 2 Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/ --info
Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/
├── [ 401820] 2018_06_19_14_50_39.info
├── [ 401821] 2018_06_19_14_50_39.mp4
├── [ 401822] analytics
│   ├── [ 401823] 2018_06_19_14_50_39.map
│   └── [ 401824] 2018_06_19_14_50_39.tlv
```

```
2018_06_19_14_50_39.info
File Edit Search Options Help
#Tue Jun 19 14:55:41 EDT 2018
CameraType=23
Source_File_Path=/storage/emulated/0/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
FPS_Drone=30
EndTimeMsec=234833
LocalFileName=2018_06_19_14_50_39
StartTimeMsec=0
FolderID_Drone=100
WhiteBalance=0
ImageDescription=
Is_HD=false
ISO=0
PixelXDimension_Drone=1920
PixelYDimension_Local=720
Video_Type=0
ExposureMode=1
PixelXDimension_Local=1280
Sync_Drone_Time=2798,21406,31080,40573,47491,51419,56131,62052,66403,72863,82480,89239,951
Version=1.0
Video_Resolution_Enum_Drone=10
FPS_local=30
CaptureDate=2018/06/19 14\:50\:39
UUID_Drone=3890597672
File_Source_Type=1
FileID_Drone=1
ProductType=38
LocationString=
PositionRelativeAlt=12.0
PositionGPSLat=39.96119532709023
FrameJumped=14
ApertureSize=560
PixelYDimension_Drone=1080
Sync_Local_Time=0,17366,25766,33833,39466,42100,45166,50000,53233,58233,66266,71566,76566
ShutterSpeed=1/100
DeviceMaker=DJI
PositionGPSAlt=12.0
PositionGPSLng=-106.2164633258356
```

Analyze detailed flight record

*.dat

Power on -> Power off

kali@kali: ~/Marvic_air 113x33

```
(kali@kali)-[~/Marvic_air]
$ tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/
├── DJIFlightRecord_2018-06-19_[14-50-34].txt
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT ←
    └── 18-06-19-02-59-05_FLY058.DAT
```

Power on: UTC time: 2018-06-19 16:47:38

1 directory, 3 files

```
(kali@kali)-[~/Marvic_air]
$ stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
Size: 4002791      Blocks: 7824      IO Block: 4096   regular file
Device: 801h/2049d Inode: 401984   Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2021-05-04 11:14:18.719351445 -0400
Modify: 2018-06-19 12:58:24.000000000 -0400 ←
Change: 2021-05-03 15:24:15.403574227 -0400
Birth: 2021-05-03 15:24:15.379562225 -0400
```

Power off: UTC time: 2018-06-19 16:58:24

~11 minutes

Metric / Imperial
Settings

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Jun 19th, 2018 12:49PM

incorrect

16:47:38

Edit

GENERAL

POWER

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WEATHER

MEDIA

Jun 19th, 2018
12:49PM (-06:00)

Plane Name
Default Mavic Air ..

Flight Air Time
05m 01s

Takeoff Battery
97% 12.6v

Landing Battery
65% 11.7v

Mavic Air
DJI DAT 1.17

Map

Satellite



Google

Map Data Terms of Use Report a map error

Total Mileage
155,605 ft

Max Distance
1,688 ft

Max Altitude
86.3 ft

Max Speed
7.65 mph

Max Bat Temp
110.8°F

Tips: 1
Warnings: 0

Download: [KML](#) [GPX \(?\)](#) [CSV](#) [Original](#)

[Metric / Imperial](#)
[Settings](#)

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GENERAL

Associated Pilots:



United States

POWER

Take Off Location:

39.964941,-106.216820
Above Sea Level: **8122.0 ft**

SENSORS

Address:

Unnamed Road, Parshall, CO 80468, USA
[Edit Address](#)

CONTROLS

Last Known Location:

39.962717,-106.217468
at log time of **12m 16s**
and altitude of **-1.5 ft**

WEATHER

Log Duration:

12m 21s (time in the air and on the ground)

Air Duration:

05m 01s (time in the air only)

[Metric / Imperial](#)
[Settings](#)

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GENERAL

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CONTROLS

Associated Drone: Default Mavic Air Drone

Type **Mavic Air**

Associated Batteries: Default Mavic Air Battery



GENERAL

POWER

SENSORS

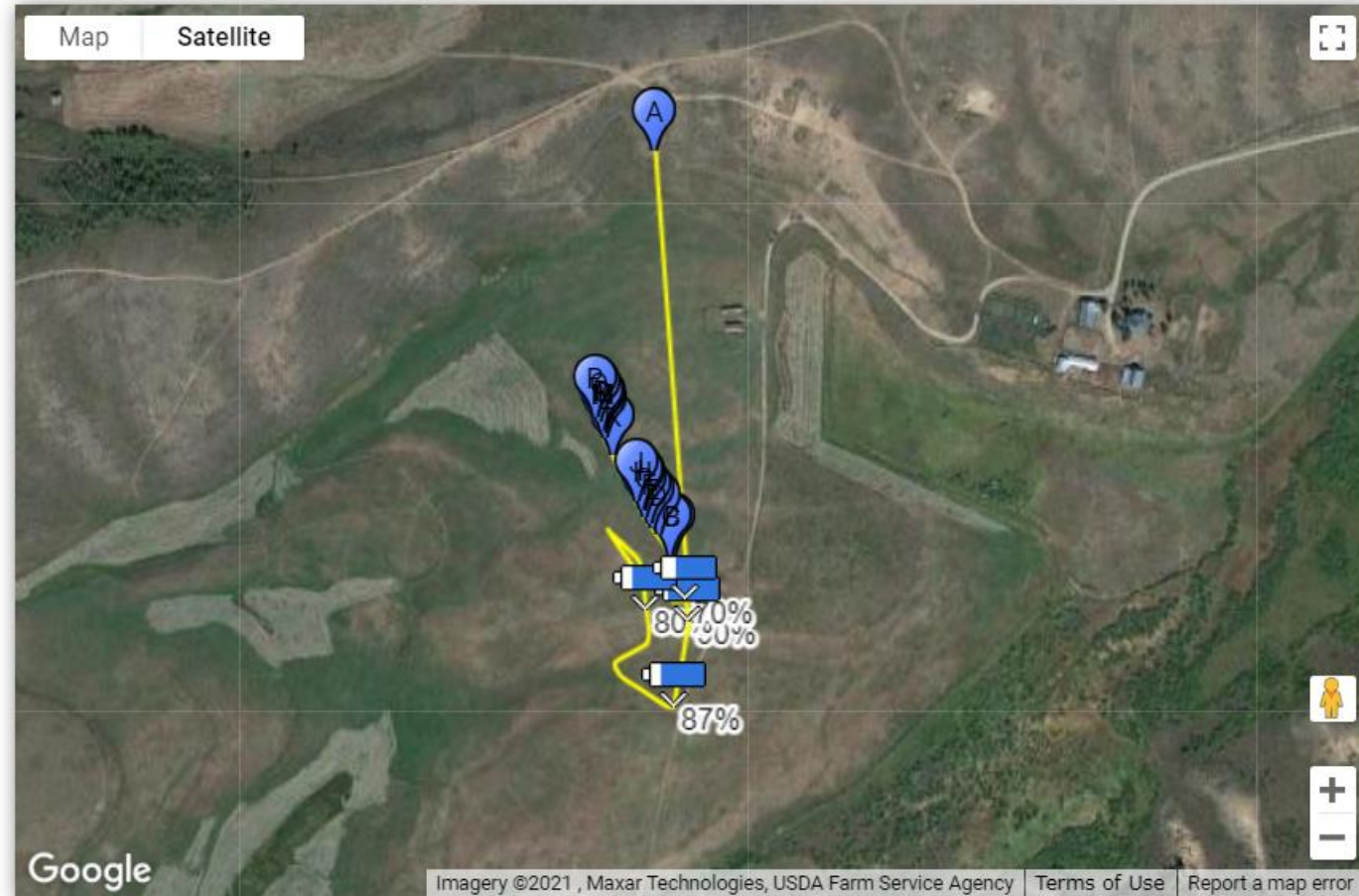
CONTROLS

WEATHER

MEDIA

HD Flight Player

Review notification highlights from the flight.
Click "HD Flight Player" to view detailed flight info.



HD Flight Player *NEW!*

[Download notifications KML](#)

Flight time	Altitude	Home Dist	Type	Notification
00m 02s	0.0 ft	0 ft	Mode	Mode changed to Manual

	Flight time	Altitude	Home Dist	Type	Notification
	00m 02s	0.0 ft	0 ft	Mode	Mode changed to Manual
	00m 02s	0.0 ft	0 ft	Tip	Log filename: FLY057.DAT
A	<u>03m 32s</u>	-8.4 ft	0 ft	Mode	Mode changed to Manual
	<u>05m 28s</u>	4.0 ft	1,431 ft		90% Battery
	<u>06m 07s</u>	65.2 ft	1,688 ft		87% Battery at maximum distance
	<u>07m 16s</u>	73.9 ft	1,393 ft		80% Battery
	<u>09m 07s</u>	-9.5 ft	1,369 ft		70% Battery
B	<u>10m 52s</u>	-1.0 ft	1,366 ft	Mode	Mode changed to Manual
C	<u>10m 52s</u>	-1.3 ft	1,366 ft	Mode	Mode changed to Manual
D	<u>11m 01s</u>	-1.5 ft	1,366 ft	Mode	Mode changed to Manual
E	<u>11m 05s</u>	-1.0 ft	1,366 ft	Mode	Mode changed to Manual
F	<u>11m 09s</u>	-1.1 ft	1,366 ft	Mode	Mode changed to Manual
G	<u>11m 13s</u>	-1.5 ft	1,366 ft	Mode	Mode changed to Manual
H	<u>11m 18s</u>	-1.0 ft	1,366 ft	Mode	Mode changed to Manual
I	<u>11m 18s</u>	-0.8 ft	1,366 ft	Mode	Mode changed to Manual
J	<u>11m 24s</u>	-1.0 ft	1,366 ft	Mode	Mode changed to Manual
K	<u>11m 53s</u>	-0.0 ft	1,366 ft	Mode	Mode changed to Manual
L	<u>12m 04s</u>	-1.0 ft	1,366 ft	Mode	Mode changed to Manual
M	<u>12m 06s</u>	-1.1 ft	1,366 ft	Mode	Mode changed to Manual
N	<u>12m 07s</u>	-1.2 ft	1,366 ft	Mode	Mode changed to Manual
O	<u>12m 08s</u>	-1.3 ft	1,366 ft	Mode	Mode changed to Manual
P	<u>12m 08s</u>	-1.2 ft	1,366 ft	Mode	Mode changed to Manual
Q	<u>12m 13s</u>	-1.4 ft	1,366 ft	Mode	Mode changed to Manual
R	<u>12m 16s</u>	-1.5 ft	842 ft	Mode	Mode changed to Manual

Analyze cached pictures

What is the location of cache pictures?

```
kali@kali: ~/Marvic_a  
$ tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/  
Android_Logical/sdcard/DJI/dji.go.v4/  
├── CACHE_IMAGE  
├── databases  
├── DJI_RECORD  
├── DJI_SPALSH  
├── editor  
├── FlightRecord  
├── LOG  
├── Package  
├── RECORD_VOICE  
├── VideoEditor  
├── videoguide  
├── visual_guide  
└── xtScreenshots
```

How many pictures contain people?



Any pictures taken on 2018-06-19?

```
kali@kali: ~/Marvic_air
(kali@kali)-[~/Marvic_air]
$ tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE ←
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots
```

```
(kali@kali)-[~/Marvic_air]
$ tree Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE
Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE
├── thumb_33177fb665068406_1523611394000.jpg
├── thumb_5be64acbef28816c_1523610990000.jpg
├── thumb_6d9f39f0ca9decc3_1523610988000.jpg
├── thumb_7f92b744bbc53b3a_1523610992000.jpg
├── thumb_dd609e61934788b3_1523611494000.jpg
└── thumb_ed8c1a297e1061b4_1524152656000.jpg
```

The sequence of pictures were taken

```
kali@kali: ~/Marvic_air 109x27

(kali@kali)-[~/Marvic_air]
$ ls --full-time -tr Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/
total 104
-rw-r--r-- 1 kali kali 19508 2018-06-19 12:47:50.000000000 -0400 thumb_7f92b744bbc53b3a_1523610992000.jpg
-rw-r--r-- 1 kali kali 12755 2018-06-19 12:47:50.000000000 -0400 thumb_6d9f39f0ca9decc3_1523610988000.jpg
-rw-r--r-- 1 kali kali 19191 2018-06-19 12:47:50.000000000 -0400 thumb_5be64acbef28816c_1523610990000.jpg
-rw-r--r-- 1 kali kali 15631 2018-06-19 12:47:52.000000000 -0400 thumb_ed8c1a297e1061b4_1524152656000.jpg
-rw-r--r-- 1 kali kali 13791 2018-06-19 12:47:52.000000000 -0400 thumb_dd609e61934788b3_1523611494000.jpg
-rw-r--r-- 1 kali kali 14224 2018-06-19 12:47:52.000000000 -0400 thumb_33177fb665068406_1523611394000.jpg
```

```
kali@kali: ~/Marvic_air 104x27

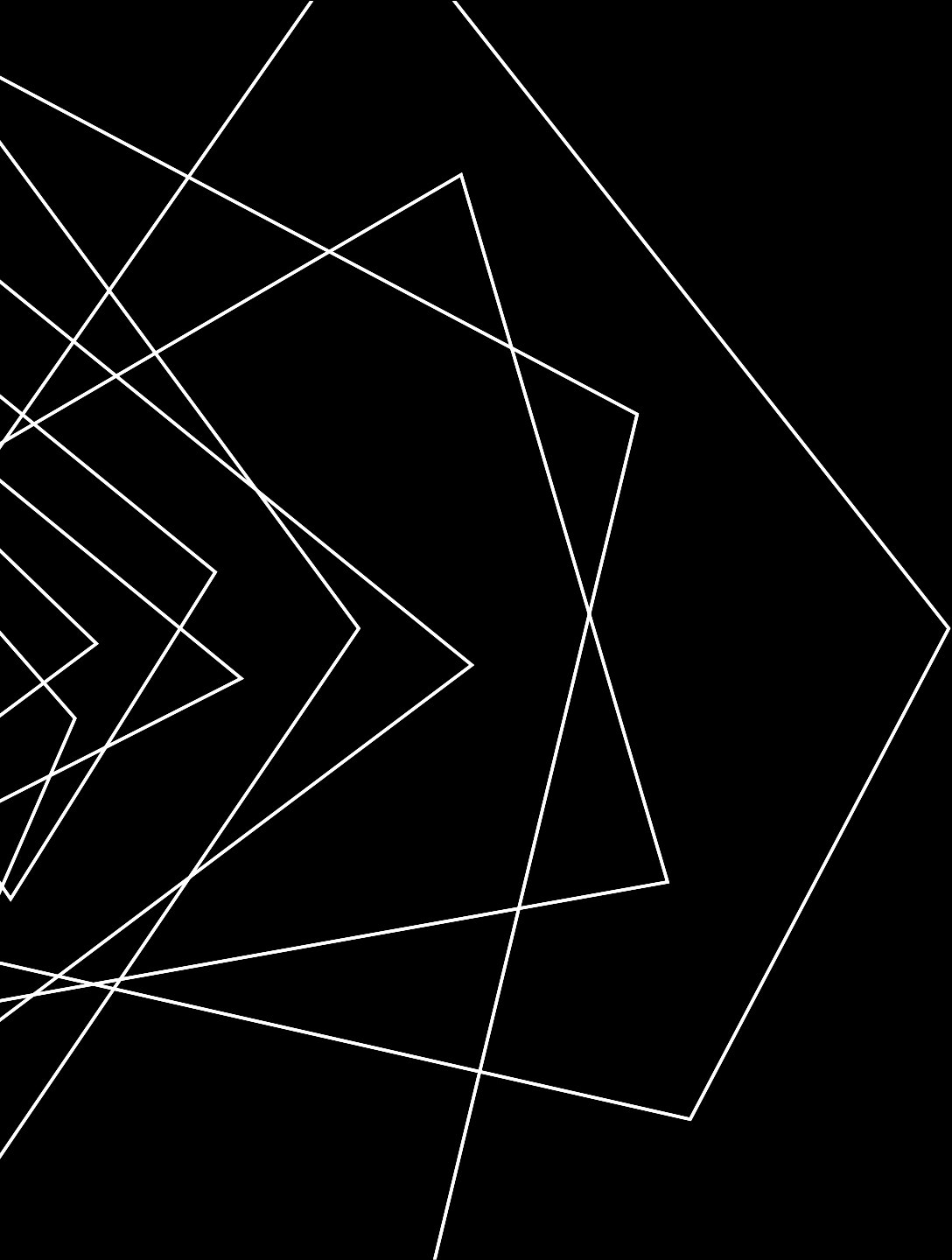
(kali@kali)-[~/Marvic_air]
$ stat Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg
File: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg
Size: 14224      Blocks: 32      IO Block: 4096   regular file
Device: 801h/2049d Inode: 401811  Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2021-05-05 15:52:29.986531740 -0400
Modify: 2018-06-19 12:47:52.000000000 -0400
Change: 2021-05-04 23:16:16.884059916 -0400
Birth: 2021-05-04 23:16:16.884059916 -0400
```


Misc.

What was the last time dji.go.v4 updated?

```
kali@kali: ~/Marvic_air 109x27  
  
(kali@kali)-[~/Marvic_air]  
$ ls --full-time -tr Android_Logical/sdcard/Android/data/dji.go.v4/cache  
total 576  
-rw-r--r-- 1 kali kali 139332 2018-06-19 11:02:20.000000000 -0400 component_upgrade_list.json  
-rw-r--r-- 1 kali kali 31686 2018-06-19 11:02:20.000000000 -0400 component_upgrade_list_br.json  
-rw-r--r-- 1 kali kali 30 2018-06-19 11:02:20.000000000 -0400 component_upgrade_date.json  
-rw-r--r-- 1 kali kali 4 2018-06-19 11:13:22.000000000 -0400 afinalCache.1  
-rw-r--r-- 1 kali kali 240032 2018-06-19 11:15:46.000000000 -0400 afinalCache.idx  
-rw-r--r-- 1 kali kali 4643 2018-06-19 11:15:46.000000000 -0400 afinalCache.0  
-rw-r--r-- 1 kali kali 139332 2018-06-19 11:36:24.000000000 -0400 list.json  
-rw-r--r-- 1 kali kali 30 2018-06-19 12:12:56.000000000 -0400 date.json  
drwxr-xr-x 2 kali kali 4096 2018-09-22 22:30:52.000000000 -0400 afinalCache  
drwxr-xr-x 2 kali kali 4096 2018-10-03 11:57:24.000000000 -0400 uil-images
```





Thank you

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